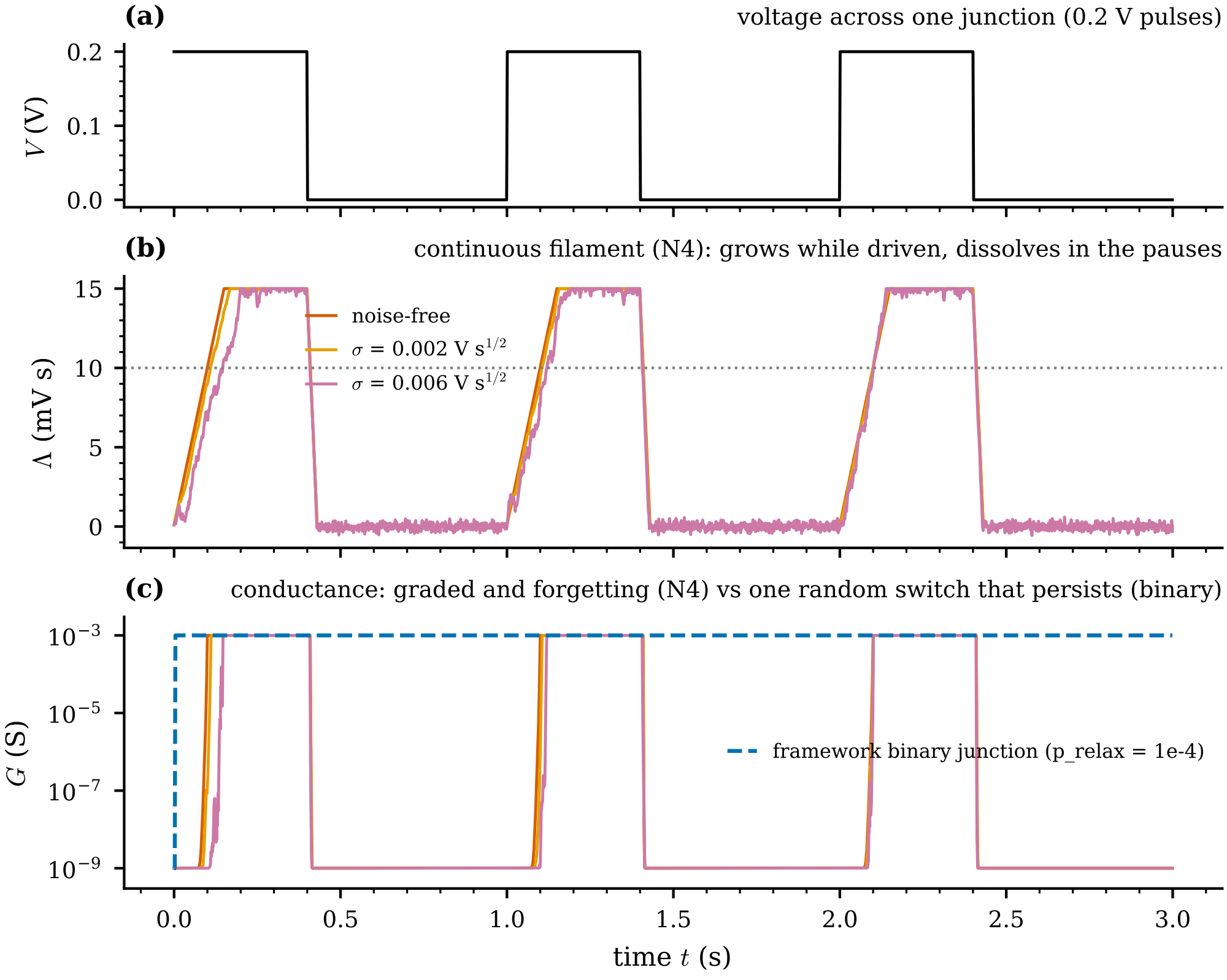

(a)
voltage across one junction (0.2 V pulses)
0.2
0.1
0.0
V (V)
(b)
continuous filament (N4): grows while driven, dissolves in the pauses
15
10
5
0
Λ (mV s)
noise-free
σ = 0.002 V s^1/2
σ = 0.006 V s^1/2
(c)
conductance: graded and forgetting (N4) vs one random switch that persists (binary)
10^−3
10^−5
10^−7
10^−9
G (S)
framework binary junction (p_relax = 1e-4)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
time t (s)